Task 5: Exploratory Data Analysis (EDA)

Dataset: Titanic passenger dataset

Objective: Use statistical summaries and visual exploration to identify patterns, trends, missing values, and unusual observations.

1. Tools and method

Python, Pandas, Matplotlib, and Seaborn were selected. The workflow checks data shape and types, uses describe() and value_counts(), examines missing data, and plots distributions, group comparisons, scatterplots, a correlation heatmap, and a pairplot.

2. Dataset overview

The standard public Titanic dataset (891 rows, 15 columns) is referenced; run the notebook online to load it..

Reference dataset checks: 342 passengers survived (38.38%); age has 177 missing entries; deck/cabin has 688 missing entries.

3. Key observations

• Survival outcomes are unevenly distributed, so a simple overall average is not enough.

• Survival rates differ by sex and passenger class; bar charts make these group differences easier to compare.

• Age has missing values, and deck/cabin information is substantially incomplete. Missingness should be assessed before modelling.

• Fare values have a wide range and high-value observations. Median and interquartile range can be useful alongside mean and standard deviation.

• Correlation heatmaps and pairplots help identify associations, but correlation does not imply causation.

4. Recommendations

1. Report missing-value counts and percentages before deciding how to treat them. 2. Compare outcomes across relevant segments rather than relying only on overall averages. 3. Investigate outliers before removing them. 4. Add a short, plain-language observation under every chart. 5. Use correlation as a starting point for investigation, not as proof of cause.

5. Conclusion

EDA combines descriptive statistics and visualizations to understand data quality, distributions, group differences, and potential relationships before formal modelling. The notebook contains reproducible code for the full analysis.

Dataset source: Seaborn public Titanic dataset — https://github.com/mwaskom/seaborn-data/blob/master/titanic.csv